

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q17-0794

Macheche 138 kV
May 21, 2018

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0794. The Interconnection Customer proposes a 71.6 MW solar photovoltaic facility interconnecting at the Macheche 138 kV substation in the West Virginia area. The request was received on May 07, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 358 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 19 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$29,431,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Low Gp Substation - Jack Mcdonough 138 KV	Artesia Country Club - Clark County 345 KV	198.1	273	6.4
Frederick - Altavista Power Station 138 KV	Lcec No 5 - Shepherdstown 138 KV	179.0	224	5.0
Lamont - Frederick 138 KV L	Lcec No. 5 - Shepherdstown 138 KV	151.7	288	5.0
Schafrath Switch - Dam No. 4 Hydro Station 138 KV Line	American - Urban 345 KV	138.9	241	51.1
Nipetown - Schafrath Switch 138 KV Line	American - Urban 345 KV L	132.6	262	51.1
Dam No. 5 Hydro Sta - Nipetown 138 KV Line	American - Urban 345 KV line	132.4	269	51.1
Lascassas - Dam No. 5 Hydro Sta 138 KV	American - Urban 345 KV	125.1	295	51.1
Lamont - University City Substation 138 KV	Clark County - Kearneyville 345 KV	122.6	291	8.8
University City - Richford 138 KV	Clark County - Kearneyville 345 KV	118.6	330	8.8
Rihford Substation - Sun City 138 KV	Clark County - Kearneyville 345 KV	117.2	345	8.8
Sun City - Peachland Substation 138 KV line	Clark County Substation - Kearneyville 345 KV	116.4	355	8.8
Lcc No. 5 - Shepherdstown 138 KV	Urban - American 345 KV	105.3	338	18.9
Cameron - Pluess 138 KV	Dam No. 4 Hydro Sta - Gravel Ridge Substation 138 KV Line	105.2	368	6.9
Arvin - Shuffleton 138 KV	Clark County - Kearneyville 345 KV	103.9	223	5.6

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Bon Aqua - Dry Gap 138 KV	Dam No. 4 Hydro Station Substation - Gravel Ridge 138 KV	102.7	813	6.0
Bon Aqua - Carl Bailey 138 KV	Dam No. 4 Hydro Station - Gravel Ridge 138 KV line	102.5	820	6.0
Carl Bailey - Otsego 138 KV	Dam No. 4 Hydro Sta - Gravel Ridge 138 KV line	102.4	831	6.0
Artesia Country Club Substation - Ross Fork 138 KV	Artesia Country Club - Air Force Base 345 KV L	102.0	765	6.4
Air Force Base 345-138 kV XFMR	Tiger Bay - Air Force Base 345 KV line	101.3	541	5.2

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Low Gap - Jack Mcdonough 138 KV line	Rebuild line with higher-capacity conductor	\$2,177,000	2% (\$51,000)
Frederick Substation - Altavista Power Station 138 KV	Rebuild line with higher-capacity conductor	\$9,387,000	22% (\$2,074,000)
Frederick - Lamont 138 KV	Rebuild line with higher-capacity conductor	\$9,941,000	22% (\$2,197,000)
Dam No. 4 Hydro Sta - Schafrath Switch 138 KV	Rebuild line with higher-capacity conductor	\$2,462,000	100% (\$2,462,000)
Schafrath Switch - Nipetown 138 KV	Rebuild line with higher-capacity conductor	\$1,663,000	100% (\$1,663,000)
Nipetown Substation - Dam No. 5 Hydro Sta 138 KV	Rebuild line with higher-capacity conductor	\$3,210,000	100% (\$3,210,000)
Lascassas - Dam No. 5 Hydro Sta 138 KV	Rebuild line with higher-capacity conductor	\$2,720,000	100% (\$2,720,000)
Lamont - University City Substation 138 KV	Rebuild line with higher-capacity conductor	\$9,457,000	42% (\$3,932,000)
University City - Richford 138 KV line	Rebuild line with higher-capacity conductor	\$4,313,000	42% (\$1,793,000)
Rihford Substation - Sun City 138 KV	Rebuild line with higher-capacity conductor	\$12,020,000	42% (\$4,997,000)
Sun City - Peachland Substation 138 KV line	Rebuild line with higher-capacity conductor	\$4,693,000	42% (\$1,951,000)
Shepherdstown - Lcec No. 5 138 KV	Reconductor line	\$816,000	20% (\$162,000)
Cameron - Pluess 138 KV	Reconductor line	\$715,000	3% (\$23,000)
Arvin - Shuffleton 138 KV	Reconductor line	\$705,000	100% (\$705,000)
Dry Gp - Bon Aqua 138 KV	Reconductor line	\$1,751,000	2% (\$35,000)
Bon Aqua - Carl Bailey 138 KV	Reconductor line	\$3,171,000	2% (\$63,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Otsego - Carl Bailey 138 KV	Reconductor line	\$2,241,000	2% (\$44,000)
Artesia Country Club - Ross Fork 138 KV Line	Reconductor line	\$2,038,000	3% (\$67,000)
Air Force Base 345-138 kV XFMR	Replace transformer with higher-rated unit	\$21,633,000	6% (\$1,282,000)
	Total allocated to project		\$29,431,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.